CS2601 Linear and Convex Optimization
Homework 1 Solution

Due: 2021.9.30

1. Let $f(\boldsymbol{x}) = 2x_1^2 + x_1x_2 + x_2^2 - 3x_1 - 5x_2$.

(a). Is $f(x)$ coercive? Hint: use $x_1x_2 \geq -(x_1^2 + x_2^2)/2$.

(b). Find the minimum and maximum of $f(\boldsymbol{x})$ over $\mathbb{R}^2$ if they exist.

Solution. (a). Note

$$f(\boldsymbol{x}) \geq 2x_1^2 - \frac{1}{2}(x_1^2 + x_2^2) + x_2^2 - 3x_1 - 5x_2 = \frac{3}{2}x_1^2 + \frac{1}{2}x_2^2 - 3x_1 - 5x_2.$$

Clearly the RHS goes to $+\infty$ as $\|\boldsymbol{x}\| \to \infty$. In fact,

$$\begin{aligned} \text{RHS} &\geq \frac{1}{2}(x_1^2 + x_2^2) - 5(|x_1| + |x_2|) \\ &\geq \frac{1}{2}(x_1^2 + x_2^2) - 5\sqrt{2(x_1^2 + x_2^2)} \\ &= \frac{1}{2}\|\boldsymbol{x}\|^2 - 5\sqrt{2}\|\boldsymbol{x}\| \to +\infty, \quad \text{as } \|\boldsymbol{x}\| \to \infty. \end{aligned}$$

So $f(\boldsymbol{x})$ is coercive.

(b). Since f is coercive, it has no maximum. Since f is continuous and coercive, it has a minimum. By the first-order necessary condition,

$$\nabla f(\boldsymbol{x}) = (4x_1 + x_2 - 3, x_1 + 2x_2 - 5)^T = 0 \implies \boldsymbol{x}^* = (\frac{1}{7}, \frac{17}{7})^T$$

Since it is the only point satisfying the necessary condition, it must be the minimum.

Note. We can verify the global minimality of $\boldsymbol{x}^*$ using the second-order Taylor expansion at $\boldsymbol{x}^*$. Recall the second-order Taylor expansion is exact for a quadratic function, so

$$f(\boldsymbol{x}) = f(\boldsymbol{x}^*) + \nabla f(\boldsymbol{x}^*)^T\boldsymbol{d} + \frac{1}{2}\boldsymbol{d}^T\nabla^2 f(\boldsymbol{x}^*)\boldsymbol{d} = f(\boldsymbol{x}^*) + \frac{1}{2}\boldsymbol{d}^T\nabla^2 f(\boldsymbol{x}^*)\boldsymbol{d}, \quad \text{where } \boldsymbol{d} = \boldsymbol{x} - \boldsymbol{x}^*.$$

The Hessian matrix

$$\nabla^2 f(\boldsymbol{x}^*) = \begin{pmatrix} 4 & 1 \\ 1 & 2 \end{pmatrix}$$

is positive definite, since $\det(4) = 4 > 0$, $\det \nabla^2 f = 7 > 0$. Therefore, $f(\boldsymbol{x}) \geq f(\boldsymbol{x}^*)$.

□

2. Logistic regression. Recall the objective function of logistic regression is the following negative log likelihood,

$$f(\boldsymbol{w}) = \sum_{i=1}^{m} \log(1 + e^{-y_i \boldsymbol{x}_i^T \boldsymbol{w}}),$$

where $(\boldsymbol{x}_i, y_i) \in \mathbb{R}^n \times \{-1, +1\}$ is the i-th data point. We have absorbed the bias term b into $\boldsymbol{w}$ by appending an extra 1 to each $\boldsymbol{x}_i$.

(a). Suppose the dataset $\{(\boldsymbol{x}_1, y_1), \ldots, (\boldsymbol{x}_m, y_m)\}$ is strictly linearly separable in the sense that there exists a $\boldsymbol{w}_0$ such that

$$y_i \boldsymbol{x}_i^T \boldsymbol{w}_0 > 0, \quad \forall i = 1, 2, \ldots, m.$$

Does f have a global minimum in this case? Explain your answer.

(b). Suppose the dataset is not linearly separable in the sense that for any $\boldsymbol{w}$, there exists an $i_0 = 1, 2, \ldots, m$ such that

$$y_{i_0} \boldsymbol{x}_{i_0}^T \boldsymbol{w} < 0.$$

Show that f has a global minimum by completing the following steps.

i) Show

$$f(\boldsymbol{w}) \geq h(\boldsymbol{w})$$

where

$$h(\boldsymbol{w}) = \max_{1 \leq i \leq m} -y_i \boldsymbol{x}_i^T \boldsymbol{w}.$$

ii) Let $S = \{\boldsymbol{w} : \|\boldsymbol{w}\| = 1\}$ be the unit sphere. Show that $h(\boldsymbol{w})$ has a global minimum $\boldsymbol{w}_0$ on S and $C \triangleq h(\boldsymbol{w}_0) > 0$. You can assume the fact that h is continuous, which can be proved by induction and the identity $\max\{a, b\} = \frac{a+b+|a-b|}{2}$.

iii) Show

$$h(\boldsymbol{w}) \geq C\|\boldsymbol{w}\|, \quad \forall \boldsymbol{w}$$

iv) Show f has a global minimum.

(c). Find $\nabla f(\boldsymbol{w})$.

Solution. (a). No, f does not have a global minimum. Suppose $\boldsymbol{w}_0$ separates the dataset. Then

$$\lim_{t \to +\infty} f(t\boldsymbol{w}_0) = \sum_{i=1}^{m} \lim_{t \to +\infty} \log(1 + e^{-t y_i \boldsymbol{x}_i^T \boldsymbol{w}_0}) = 0$$

On the other hand $f(\boldsymbol{w}) > 0$ for all $\boldsymbol{w} \in \mathbb{R}^n$. So $\inf_{\boldsymbol{w}} f(\boldsymbol{w}) = 0$, but it is not achievable.

(b). i) Note

$$\log(1 + e^{-y_i \boldsymbol{x}_i^T \boldsymbol{w}}) \geq \log e^{-y_i \boldsymbol{x}_i^T \boldsymbol{w}} = -y_i \boldsymbol{x}_i^T \boldsymbol{w}$$

so

$$f(\boldsymbol{w}) = \sum_{i=1}^{m} \log(1 + e^{-y_i \boldsymbol{x}_i^T \boldsymbol{w}}) \geq \max_{1 \leq i \leq m} \log(1 + e^{-y_i \boldsymbol{x}_i^T \boldsymbol{w}}) \geq \max_{1 \leq i \leq m} -y_i \boldsymbol{x}_i^T \boldsymbol{w} = h(\boldsymbol{w}).$$

ii) Since h is continuous and S is compact, by the Extreme Value Theorem, h has a global minimum $\boldsymbol{w}_0$ on S. By assumption, there exists some i_0 such that $y_{i_0}\boldsymbol{x}_{i_0}^T\boldsymbol{w}_0 < 0$, so

$$C = h(\boldsymbol{w}_0) = \max_{1\le i\le m} -y_i\boldsymbol{x}_i^T\boldsymbol{w}_0 \ge -y_{i_0}\boldsymbol{x}_{i_0}^T\boldsymbol{w}_0 > 0.$$

iii) If $\boldsymbol{w} = 0$, $h(\boldsymbol{w}) = 0 = C\|\boldsymbol{w}\|$. Suppose $\boldsymbol{w} \neq 0$ and let $\hat{\boldsymbol{w}} = \frac{\boldsymbol{w}}{\|\boldsymbol{w}\|}$. Since $\|\hat{\boldsymbol{w}}\| = 1$, $h(\hat{\boldsymbol{w}}) \ge C$. Thus

$$h(\boldsymbol{w}) = h(\|\boldsymbol{w}\|\hat{\boldsymbol{w}}) = \|\boldsymbol{w}\|h(\hat{\boldsymbol{w}}) \ge C\|\boldsymbol{w}\|.$$

iv) Note f is coercive, since $f(\boldsymbol{w}) \ge h(\boldsymbol{w}) \ge C\|\boldsymbol{w}\|$. Since it is also continuous, f must have a global minimum.

(c).

$$\nabla f(\boldsymbol{w}) = \sum_{i=1}^m \frac{e^{-y_i\boldsymbol{x}_i^T\boldsymbol{w}}}{1+e^{-y_i\boldsymbol{x}_i^T\boldsymbol{w}}}(-y_i\boldsymbol{x}_i)$$

□

3. Taylor expansion. Suppose $f : \mathbb{R}^n \to \mathbb{R}$ is twice continuously differentiable.

(a). Show that the following first-order Taylor expansion with Lagrange remainder,

$$f(\boldsymbol{x}+\boldsymbol{d}) = f(\boldsymbol{x}) + \nabla f(\boldsymbol{x})^T\boldsymbol{d} + \frac{1}{2}\boldsymbol{d}^T\nabla^2 f(\boldsymbol{x}+t\boldsymbol{d})\boldsymbol{d}$$

for some $t \in (0,1)$. You can assume the same expansion for univariate functions.

(b). Recall the integral of a vector valued function $\boldsymbol{g} : \mathbb{R} \to \mathbb{R}^n$ is defined component-wise, i.e. for $\boldsymbol{g}(t) = (g_1(t), g_2(t), \ldots, g_n(t))^T$,

$$\int \boldsymbol{g}(t)dt = \begin{pmatrix} \int g_1(t)dt \\ \int g_2(t)dt \\ \vdots \\ \int g_n(t)dt \end{pmatrix}$$

Show

$$\nabla f(\boldsymbol{x}+\boldsymbol{d}) = \nabla f(\boldsymbol{x}) + \int_0^1 \nabla^2 f(\boldsymbol{x}+t\boldsymbol{d})\boldsymbol{d}\, dt$$

Hint: Apply the Newton-Leibniz formula to $\boldsymbol{g}(t) = \nabla f(\boldsymbol{x}+t\boldsymbol{d})$.

Proof. (a). Let $g(t) = f(\boldsymbol{x}+t\boldsymbol{d})$. By the first-order Taylor expansion for g,

$$g(1) = g(0) + g'(0) + \frac{1}{2}g''(t)$$

for some $t \in (0,1)$. By the chain rule, $g'(0) = \nabla f(\boldsymbol{x})^T\boldsymbol{d}$ and $g''(t) = \boldsymbol{d}^T\nabla^2 f(\boldsymbol{x}+t\boldsymbol{d})\boldsymbol{d}$, so

$$f(\boldsymbol{x}+\boldsymbol{d}) = f(\boldsymbol{x}) + \nabla f(\boldsymbol{x})^T\boldsymbol{d} + \frac{1}{2}\boldsymbol{d}^T\nabla^2 f(\boldsymbol{x}+t\boldsymbol{d})\boldsymbol{d}.$$

(b). Let $\boldsymbol{g}(t) = \nabla f(\boldsymbol{x} + t\boldsymbol{d})$. By the Newton-Leibniz formula,

$$\boldsymbol{g}(1) = \boldsymbol{g}(0) + \int_0^1 \boldsymbol{g}'(t)dt$$

By the chain rule, $\boldsymbol{g}'(t) = \nabla^2 f(\boldsymbol{x} + t\boldsymbol{d})\boldsymbol{d}$, so

$$\nabla f(\boldsymbol{x} + \boldsymbol{d}) = \nabla f(\boldsymbol{x}) + \int_0^1 \nabla^2 f(\boldsymbol{x} + t\boldsymbol{d})\boldsymbol{d}\, dt$$

□

4. Are the following matrices positive definite, positive semidefinite, negative definite, negative semidefinite, or indefinite?

$$A = \begin{pmatrix} 6 & 2 & 0 \\ 2 & 5 & -2 \\ 0 & -2 & 4 \end{pmatrix}, \quad B = \begin{pmatrix} 1 & 2 & 0 \\ 2 & 2 & 1 \\ 0 & 1 & -3 \end{pmatrix}, \quad C = \begin{pmatrix} 2 & -1 & -1 \\ -1 & 2 & -1 \\ -1 & -1 & 2 \end{pmatrix}$$

Proof. We apply the determinant test.

(a). Check the leading principal minors,

$$\det(6) = 6 > 0, \quad \det \begin{pmatrix} 6 & 2 \\ 2 & 5 \end{pmatrix} = 26 > 0, \quad \det A = 80 > 0$$

so A is positive definite. Alternatively, you can check the eigenvalues are $2, 5, 8$, all positive.

(b). Check the leading principal minors,

$$\det(1) = 1 > 0, \quad \det \begin{pmatrix} 1 & 2 \\ 2 & 2 \end{pmatrix} = -2 < 0, \quad \det B = 5 > 0$$

B is not positive semidefinite. Similarly, $-B$ is not positive semidefinite, so B is indefinite. Alternatively, you can check the eigenvalues are $3.65616639, -0.42297269, -3.23319369$.

(c). Check the leading principal minors,

$$\det(2) = 2 > 0, \quad \det \begin{pmatrix} 2 & -1 \\ -1 & 2 \end{pmatrix} = 3 > 0, \quad \det C = 0$$

It is not positive definite. In order to conclude that it is positive semidefinite, we need to check the other principal minors. Luckily, all first order principal minors are the same. So are all second order principal minors. Therefore, C is positive semidefinite.

You can also check the eigenvalues. But when you use software, watch out for numerical errors due to the finite precision of your machine, especially when you get some numbers with extremely small absolute values. For example, if you use `numpy.linalg.eig`, the returned eigenvalues are

$$3.00000000e + 00, -4.44089210e - 16, 3.00000000e + 00$$

It is not hard to guess they should be $3, 0, 3$, and you should verify this by showing the characteristic polynomial is

$$\det(\lambda I - C) = \lambda(\lambda - 3)^2.$$

$\square$